

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COLLEGE OF ENGINEERING GUINDY

ANNA UNIVERSITY

INDUSTRY ORIENTED COURSE

Course Title	Scalable Enterprise Architectural Deployments of Agentic AI Solutions
Programme & Branch	B.E CSE
Year & Semester	II Year & IV Semester
Regulation	R 2023

1. Brief Introduction about the Expert



Ms. Chandravadhana TK is a **Senior AI/ML Enterprise Architect** with over **19 years of industry experience** in Enterprise Architecture, AI/ML, Generative/Agentic AI, Cloud Computing, and Digital Transformation. She is currently working with **Optum.AI (UnitedHealth Group)**, where she architects enterprise-scale AI solutions for the healthcare domain.

Throughout her career with organizations including **Atos (American Express, Northern Trust), SmarTek (Samsung), Honeywell, and General Electric**, she has successfully delivered large-scale enterprise modernization programs, cloud-native architectures, global Generative AI and Agentic AI platforms, and mission-critical digital transformation initiatives. She established a global **Generative AI Center of Excellence**, mentored over **250 engineers** across multiple geographies, organized organization-wide AI hackathons, and developed reusable AI frameworks that generated multi-million-dollar business value.

Her expertise spans **Agentic AI, Large Language Models (LLMs), Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Enterprise AI Architecture, MLOps, Azure AI, AWS, OpenShift, Enterprise Integration, and Responsible AI Governance**. She has architected and deployed production-grade intelligent applications, including **clinical agents, contract intelligence agents, autonomous support agents, research assistants, document intelligence platforms, and advanced knowledge retrieval systems** using leading Agentic AI frameworks.

Ms. Chandravadhana holds an **Executive General Management Programme (EGMP)** from **IIM Bangalore**, an **M.E. in Software Engineering** from **CEG, Anna University (Guindy)**, and is currently pursuing a **Ph.D. at Anna University (Guindy)**. She holds multiple professional certifications, including **PMP®, AWS Certified Cloud Practitioner, Microsoft Azure AI Engineer, ITIL®, Six Sigma Green Belt, Red Hat Cloud Professional, and UiPath RPA Developer**. She also serves as a **Board of Studies Member at Vel Tech University** and actively mentors professionals through industry-oriented AI workshops, hackathons, and enterprise training programs.

2. Title of the Course: Scalable Enterprise Architectural Deployments of Agentic AI Solutions

3. Pre-requisite

- Basic Python Programming
- Fundamentals of Machine Learning and Generative AI
- API Development Concepts
- Cloud Computing Fundamentals (AWS/Azure/GCP)
- Microservices and Containerization Basics
- Understanding of Enterprise Software Architecture

4. Lesson Plan

S.No	Module	Hours
1	Foundations of Agentic AI & Enterprise Architecture	2
2	Enterprise AI Agent Design Patterns	2
3	LLMs, Tools, Memory & Knowledge Integration	2
4	Multi-Agent Systems & Orchestration Frameworks	2
5	Enterprise Integration Architecture	2
6	Scalability, Deployment & Cloud Architecture	2
7	Security, Governance & Responsible AI	1.5
8	Observability, Monitoring & Optimization	1
9	Capstone Project: Enterprise Agentic AI Deployment	0.5
Total		15 Hours

5. Tentative Schedule

Date	Session	Topics Covered
Day 1	Session 1	Module 1: Foundations of Agentic AI & Enterprise Architecture (2 Hours) Topics Covered • Evolution from AI Assistants to AI Agents • Agentic AI Concepts and Terminologies • Enterprise AI Adoption Trends • Autonomous vs Human-in-the-Loop Systems • Enterprise Architecture Fundamentals • AI Reference Architecture Layers • Agent Lifecycle Management Hands-on • Exploring Agentic AI Use Cases across Industries
	Session 2	Module 2: Enterprise AI Agent Design Patterns (2 Hours) Topics Covered • Single-Agent Architecture • Supervisor-Agent Pattern • Planner-Executor Pattern • Reflection and Self-Correction Pattern • Human Approval Workflows • Event-Driven Agent Architecture • Workflow-based Agents Hands-on • Designing Enterprise Agent Workflows
Day 2	Session 3	Module 3: LLMs, Tools, Memory & Knowledge Integration (2 Hours)

Date	Session	Topics Covered
		Topics Covered - Enterprise LLM Selection Strategy - Prompt Engineering for Agents - Tool Calling and Function Invocation - Retrieval Augmented Generation (RAG) - Vector Databases - Long-term and Short-term Memory - Knowledge Graph Integration - Enterprise Knowledge Repositories Hands-on - Building an Agent with Tool Calling and RAG
	Session 4	Module 4: Multi-Agent Systems & Orchestration Frameworks (2 Hours) Topics Covered - Multi-Agent Collaboration Models - Agent Communication Protocols - Task Planning and Delegation - Agent Orchestration Frameworks - CrewAI Architecture - LangGraph Fundamentals - Microsoft AutoGen - Agent-to-Agent (A2A) Communication Hands-on - Creating a Multi-Agent Business Workflow
Day 3	Session 5	Module 5: Enterprise Integration Architecture (2 Hours) Topics Covered - Integrating Agents with Enterprise Systems - APIs and Microservices - ERP Integration - CRM Integration - SQL and NoSQL Databases - Event Streaming Platforms - Enterprise Service Bus (ESB) - Enterprise Data Governance Hands-on - Agent Integration with Enterprise Database and APIs
	Session 6	Module 6: Scalability, Deployment & Cloud Architecture (2 Hours) Topics Covered - Cloud-Native AI Architecture - Kubernetes for Agentic AI - Containerization using Docker - Distributed Agent Deployment - Serverless AI Patterns - Load Balancing Strategies - High Availability and Fault Tolerance - Cost Optimization Strategies Hands-on - Deploying Agents on Kubernetes
Day 4	Session 7	Module 7: Security, Governance & Responsible AI (1.5 Hours) Topics Covered - Enterprise AI Security Frameworks - Authentication and Authorization - Secrets Management - AI Governance Frameworks - Regulatory Compliance - Data Privacy and Protection

Date	Session	Topics Covered
		- Responsible AI Principles - Risk Management Hands-on - Implementing Secure Agent Access Controls
	Session 8	Module 8: Observability, Monitoring & Optimization (1 Hour) Topics Covered - Agent Monitoring Metrics - Tracing Agent Actions - Observability Frameworks - Performance Benchmarking - Cost Monitoring - Agent Evaluation Techniques - Continuous Improvement Loops Hands-on - Building Agent Monitoring Dashboards
	Session 9	Module 9: Capstone Project – Enterprise Agentic AI Deployment (0.5 Hour) Project Scenario Enterprise AI Copilot for Operations Management Participants will build a scalable Agentic AI solution that: - Accepts natural language business queries - Retrieves information from enterprise databases - Uses multiple collaborating agents - Generates insights and recommendations - Implements governance and security controls - Deploys on cloud-native infrastructure Deliverables - Architecture Diagram - Agent Workflow Design - Deployment Strategy - Security Model - Monitoring Dashboard Design